Coronary Heart Disease Prediction on NHANES: Data Audit and Model Revalidation

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Summary

The original analysis counted 15,103 labeled rows as participants, but they belong to only 5,751 adults, and that duplication inflated two of three models. Merging the medications file, which has one row per prescription, repeated each participant once per medication.

Because copies of the same person landed in both the training and test sets, random forest reached an AUC of 0.998 and KNN 0.987. Re-evaluated with one row per person and 5-fold cross-validation, their AUCs fall to 0.858 and 0.745. Logistic regression, which cannot memorize individual rows, barely changed (0.820 to 0.859).

Background and data

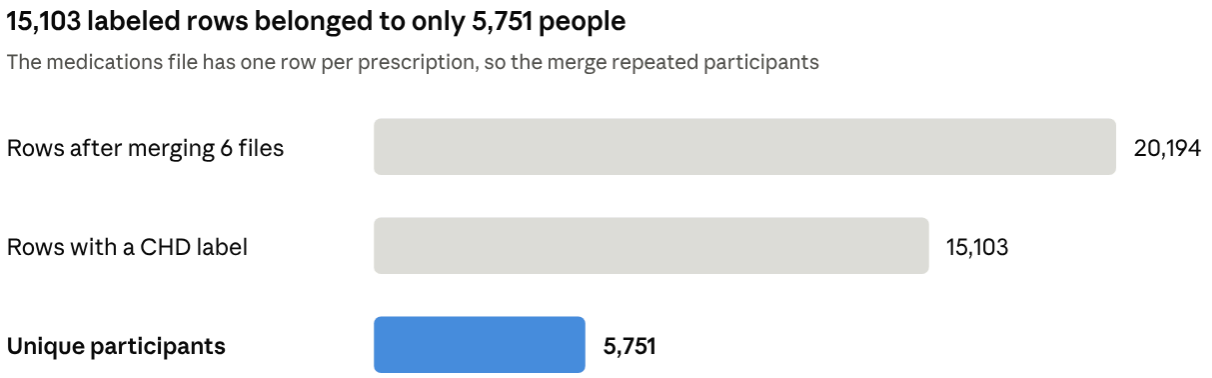
The project replicated a published PSO-ANN model that predicts coronary heart disease (CHD) from 19 routinely collected variables. It used NHANES 2013-2014 from the CDC, a survey of the U.S. population that combines interviews, physical exams and lab tests.

Item	Detail
Source files	demographic, diet, examination, labs, medications, questionnaire (joined on participant ID SEQN)
Outcome	MCQ160C: ever told by a doctor you had coronary heart disease (asked of adults 20+)
Demographics	Sex, age
Exam	Systolic and diastolic blood pressure (first reading)
Labs	Total cholesterol, HDL, triglycerides, glucose, HbA1c, creatinine, uric acid, LDH, platelet count, red-cell distribution width
Questionnaire	Vigorous work, moderate work, diabetes (DIQ010), family history of diabetes (MCQ300C), stroke history (MCQ160F)

The original preprocessing imputed missing values (mean for numeric, mode for categorical), standardized numeric variables and one-hot encoded categories. It then made a stratified 70/30 train/test split and applied SMOTE to the training set.

Data audit

The merged file repeated most participants, so the analysis cohort is 5,751 adults, not 15,103.



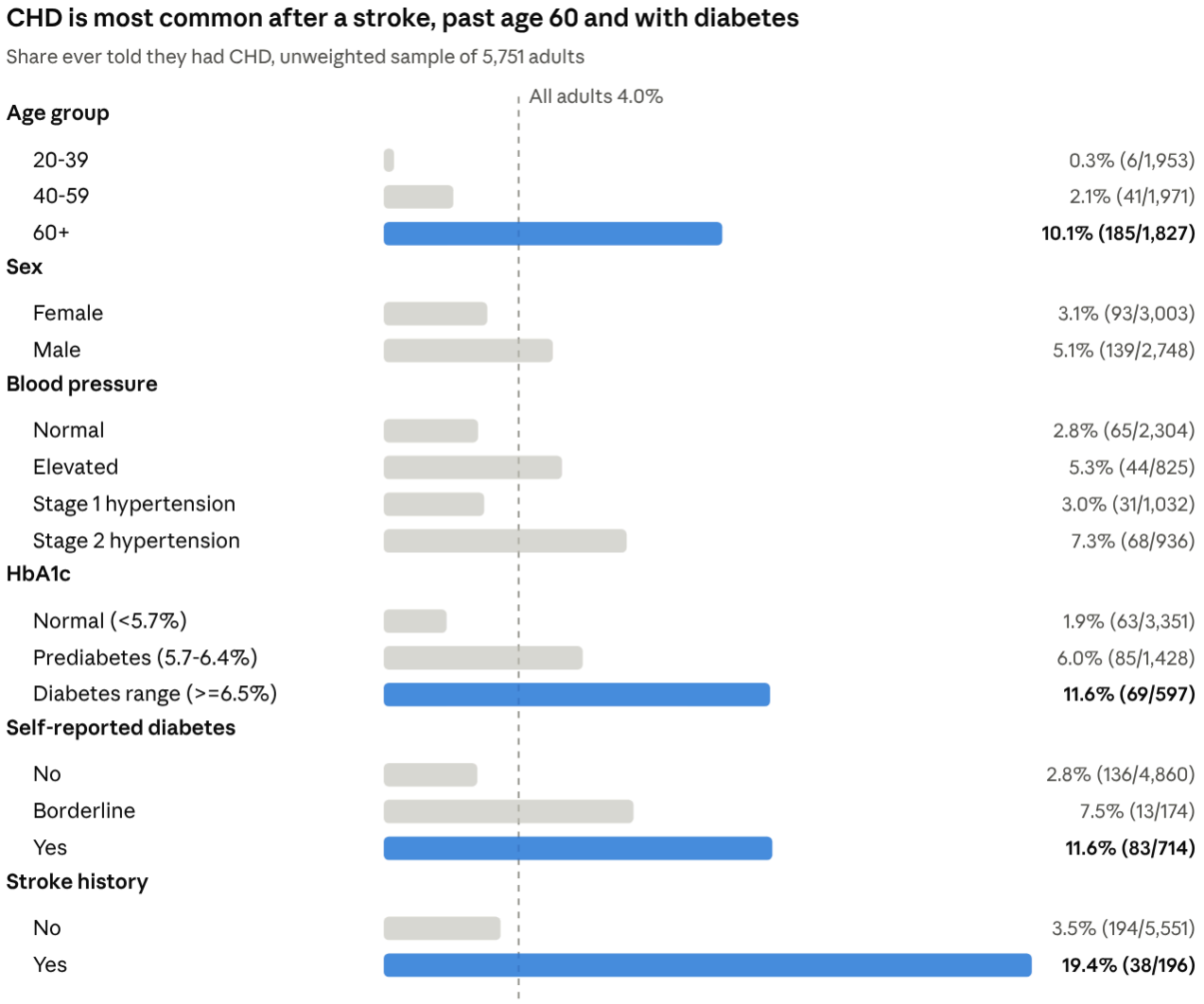
selected_features.csv from the BIOL 485 project · NHANES 2013-2014

Only 3,311 participants appear once; the other 2,440 appear between 2 and 23 times, about 2.6 rows per person on average. Duplicates are exact copies on every analysis variable, which is why a row-level split puts the same person on both sides.

In the deduplicated cohort, triglycerides are missing for 55.7% of participants, blood pressure for 11.4%, the chemistry panel for about 7.7% and the blood count for 6.5%. Sex, age and the questionnaire items are almost complete.

Cohort description

Of 5,751 adults aged 20 to 80 (median age 48; 3,003 women, 2,748 men), 232 (4.0%) reported a CHD diagnosis.

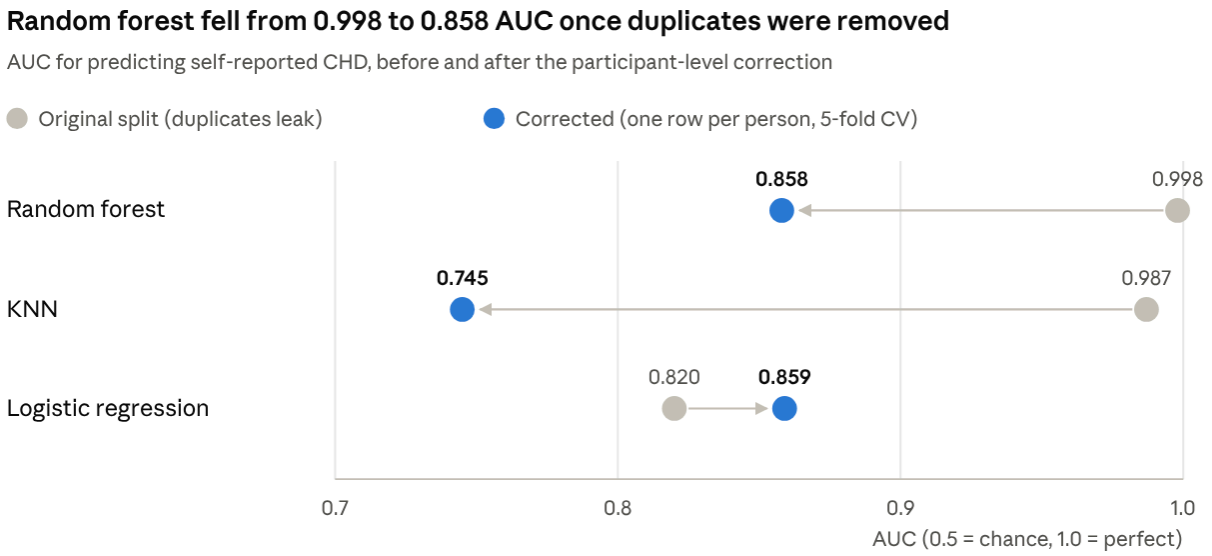


nhanes_chd_cohort.csv · 5,751 adults, NHANES 2013-2014, unweighted

Prevalence rises from 0.3% under 40 to 10.1% at 60 and over, and is about four times higher with HbA1c in the diabetes range than with normal HbA1c. Blood pressure does not rise in step: stage 1 hypertension (3.0%) sits below the elevated group (5.3%), likely because treated patients have lowered pressure. Participants with missing or unknown values are left out of each group.

Model revalidation

Once each person appears only once, the flexible models lose most of their apparent skill, and logistic regression becomes the best model.



model_comparison.csv · original test set n = 4,531 rows; corrected 5-fold CV on 5,751 adults

In the original test set, 75% of rows (98% of CHD rows) had an exact copy in the SMOTE-balanced training set, so random forest and KNN were matching people they had already seen. The corrected setup keeps one row per participant, uses stratified 5-fold cross-validation, fits imputation and scaling inside each fold, and uses class weighting instead of SMOTE.

Model (corrected)	Accuracy	Sensitivity	Specificity
Logistic regression	0.779	0.789	0.779
Random forest	0.948	0.155	0.981
KNN	0.735	0.694	0.737

Random forest's 0.948 accuracy is misleading: at a 0.5 threshold it finds only 15.5% of CHD cases, close to always predicting "no CHD" in a 4% outcome. Logistic regression finds 78.9% of cases while correctly clearing 77.9% of people without CHD. KNN uses the outcome rate (4.0%) as its threshold, because at its default 0.5 cutoff it flags no CHD cases at all (sensitivity 0).

Limitations and next steps

The corrected models are a fair internal check, but four limits remain before any number here generalizes.

- **No survey weights.** NHANES uses a complex sampling design, so prevalences here describe this sample, not U.S. adults. Weighted estimates need WTMEC2YR and the design variables (SDMVPSU, SDMVSTRA).

- **Cross-sectional, self-reported outcome.** CHD is a past diagnosis, so lab values were measured after it. Lower cholesterol among people with CHD (5.1% CHD prevalence in the desirable group vs. 2.2% in the high group) likely reflects statin treatment, not protection.
- **A mislabeled variable.** The original feature "Blood-Rel Stroke" uses MCQ160F, which asks about the participant's own stroke, not a relative's. It is relabeled StrokeHistory here.
- **The replication gap stands.** The paper reported about 0.97 accuracy; with a 4% outcome rate, accuracy is a weak target, and AUC, sensitivity and specificity are better guides.

Next steps:

1. Rerun the analysis with survey weights and report weighted prevalences.
2. Apply SMOTE inside each cross-validation fold rather than before splitting, and compare it with class weighting.
3. Tune the decision threshold for sensitivity, since missing a CHD case costs more than a false alarm in screening.
4. Test the PSO-ANN under the corrected setup to see how much of the paper's 0.97 survives.